

Report on the Tampo Printer

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Abstract

I manufactured Iran's first electric tabletop pad printing machine at this company. While previous models were either pneumatic or floor-standing units, I developed this electric, tabletop version.

The machine offers exceptional precision for very fine, small-scale printing, and virtually any design can be processed using a printing plate.

Additionally, the user can interact with the device using a character LCD, buttons, and a rotary knob to set the forward and backward limits of the pad for picking up the design from the cliché.

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Chapter 1: Introduction to the Project Overview

This project involves designing a command and control circuit board for a mechanical assembly that functions as a pad printing machine.

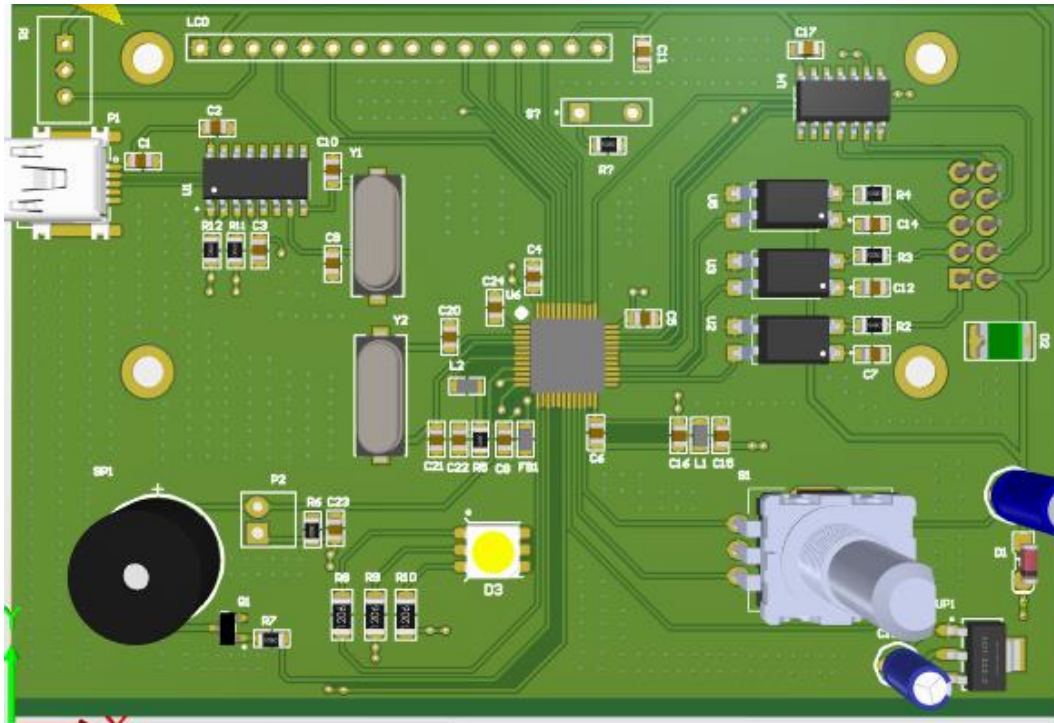
The machine operates using two stepper motors to drive the Y and Z axes; these motors are controlled via TB6600 drivers. Once the motion control is configured, the next step is to adjust the pad mounted on the Z-axis. This pad picks up the design from the printing plate (cliché) and transfers it onto the product; consequently, the pad's precise positioning and the application of appropriate pressure are critical for successfully picking up the design. The Z-axis height can also be adjusted by the user to facilitate transferring the design from the pad onto the product; these adjustments are controlled via a rotary encoder.

A 16x2 character LCD with associated menu screens has been designed to facilitate user interaction, inputting values, and configuring settings.

The print count and device code are stored in flash memory.

Chapter 2 - Introduction to the different parts of hardware

User-side equipment



The encoder button, buzzer, RGB LED, and LCD are provided for user interaction; selected values are saved using the settings button. The buzzer and RGB LED also indicate the device's overall status—whether it is running or stopped.

LCD Operating Principles

The LCD interface consists of several screens.

The main screen displays the print stroke count, pressure, and machine speed during operation.

Access to the menu screens is available after stopping the machine. Within the menu, you can navigate between options by rotating the encoder and make selections using the button; options include HOME, MIXING, SETTING, and FACTORY RESET.

The SETTING section contains parameters for pad-idle, pad-front, cup-front, speed, acceleration, and more.

Working principles of the Motors

Two stepper motors are installed to drive the device along two axes. These are controlled by two industrial 4-amp TB6600 drivers, with the microcontroller actuating the motors by sending PWM signals to the drivers' pulse inputs. The motion involves acceleration, and the user can adjust the pressure of the pad—controlled by the Z-axis motor—meaning the motor's power is also adjustable based on the generated pulses.

We have installed two microswitches at the ends of both axes to detect the device's position—or, in other words, to "home" the machine—thereby allowing the number of motor revolutions required for a full movement to be calculated.

The switches are connected to the microcontroller via an optocoupler to provide a degree of isolation from mechanical and switching noise.

Chapter 3 – Programming

To program the board, we used the CH340 IC, establishing communication with the PC via UART.

Chapter 4 – Challenges

Buzzer control

We envisioned two methods for controlling the buzzer: one via software—for instance, triggering it whenever the user rotates the rotary encoder by one step or presses its button—and the other via hardware, such that the buzzer sounds when the "Stop" button is pressed. This was implemented using two microcontroller pins: one connected to the switch and the other to the buzzer's negative terminal. This setup allows the button press to be detected via one pin while the action is executed through the other.

Power Board

The board described so far is known as the "panel board"; it houses the LCD and buttons and is installed within the panel housing to facilitate user interaction.

It was also essential to protect the microcontroller from power supply noise and potential hazards, as accidental connection of the input power by workers during operation could cause the entire board to burn out.

Consequently, a separate board—the "power board"—was developed. This board features a regulator that converts 32V DC to 5V, along with various connectors (such as ATX, XH, and IDC types) for connecting motor wires and switches, as well as for interfacing with the panel board.

